

SECTION - A

There are **FOUR** questions in this section. Answer **Q. No. 1 (compulsory)** and any other **TWO** from the rest.

1. ✓ (a) You are asked to merge two singly linked lists whose elements are sorted in ascending order. The output list will also be sorted in ascending order. **Sketch** an algorithm to solve the problem. **Analyze** that your algorithm run in linear time on the number of elements of the output list. (18)
- (b) You are given two linked lists of unsorted elements. **Investigate** whether you can **develop** a constant time algorithm to concatenate them if both of the lists are (i) singly linked lists (SLL), (ii) double linked list (DLL), (iii) circular SLL, (iv) circular DLL. If yes, **sketch** the algorithm and **examine** its correctness, otherwise **demonstrate** why a constant time algorithm is not possible. (17)
2. ✓ (a) **Show** the binary search tree (BST) that results from inserting the values 15, 20, 25, 18, 16, 5, and 7 (in that order). **Write** the enumerations of the tree that result from pre-order, in-order and post-order traversal of the tree. **Write** a recursive algorithm that takes as input the pointer to the root of a BST and a value K , and returns true if the value K appears in the tree and false otherwise. **Modify** the algorithm if the tree is a generalized binary tree. In both cases, analyze the time complexity of your algorithms. (25)
- (b) **Analyze** the maximum and minimum of numbers of nodes in a binary tree of height h . (10)
3. ✓ (a) Assume you are given a set of integer numbers. **Design** a data structure and associated algorithms that support finding minimum and maximum of the numbers in constant time and insertion, deletion of maximum, and deletion of minimum in $O(\log n)$ time. Hints: use two different heaps on the same set of numbers; however remember their mutual dependencies. (15)
- (b) Using a stack, construct an **iterative (non-recursive)** algorithm of depth first search (DFS) of a graph. **Show** how the state of the stack changes over time using the graph given in Fig. for Q. No. 3(b) assuming that the graph is represented using an adjacency list and your algorithm considers the vertices in alphabetical order. **Analyze** the complexity of your algorithm. **Modify** the algorithm so that it prints the type of each edge it encounters. (20)

Contd P/2

CSE 105

Contd ... Q. No. 3(b)

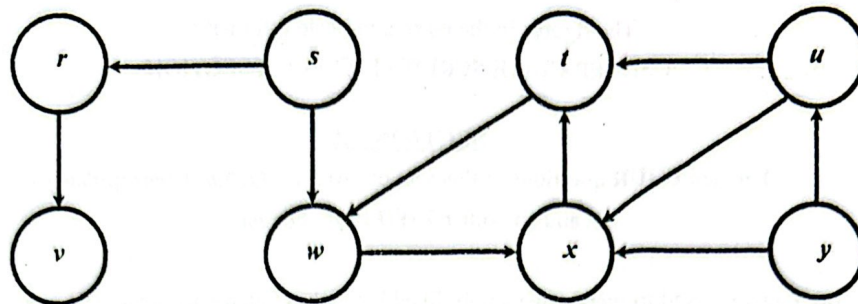


Fig. for Q. No. 3(b)

4. (a) Assume that a string contains only left and right parentheses which may or may not be balanced and properly nested. For example, the string "((()())())" contains properly nested pairs of parentheses, but the string ")()(" does not, and the string ")())" does not contain properly matching parentheses. Using a **stack**, construct an algorithm that **returns true** if a string contains properly nested and balanced parentheses, and **false otherwise**. (25)

Modify the algorithm that returns the position in the string of the first offending parentheses if the string is not properly nested and balanced. That is, if an excess right parentheses is found, **return its position**; if there are too many left parentheses, **return the position** of the first excess left parentheses. **Return - 1** if the string is properly balanced and nested. Assume indices start from 0.

- (b) Given an integer k and a queue of integers, **how do you reverse** the order of the first k elements of the queue, leaving the other elements in the same relative order? For example, if $k = 4$ and queue has the elements [10, 20, 30, 40, 50, 60, 70, 80, 90]; the output should be [40, 30, 20, 10, 50, 60, 70, 80, 90]. (10)

SECTION - B

There are **FOUR** questions in this section. **Question 5 is mandatory.**

Answer any **TWO** questions from questions 6-8.

5. (a) Assume that you have obtained a gold bar of integer length n . You plan to cut it into smaller pieces and sell them to the market. Further assume that, for any integer $i \leq n$ you know the profit you will obtain if you sell a gold bar of length i to the current market. You are now facing the problem of finding the maximum profit that you can obtain by cutting the bar into smaller pieces and selling them. (7+8=15)

(i) Using the cut-and-paste method, **formulate** an argument to show that the problem of obtaining the maximum profit by cutting this gold bar into smaller pieces (and selling them) has optimal substructure property.

(ii) **Construct** a recursion tree diagram to show that the problem has overlapping subproblem property.

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CSE 105

Contd ... Q. No. 5

(b) Analyze the following recurrences using the Master Theorem,

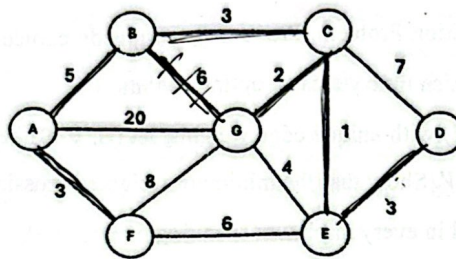
(5+5=10)

(i) $T(n) = 2T\left(\frac{n}{2}\right) + n \log n$

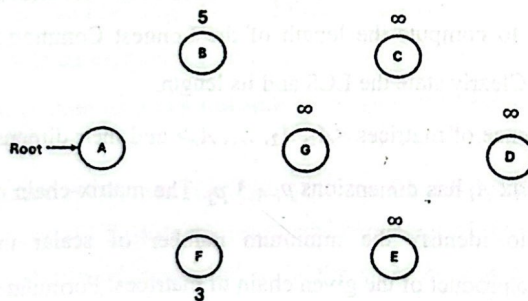
(ii) $T(n) = 2T\left(\frac{n}{2}\right) + n^2$

(c) Apply Prim's algorithm to find the minimum spanning tree of the following graph.

(10)



Start with node A as the root node. At each step, show the intermediate MST (i.e., the partial tree) and show the estimated cost to connect each non-visited node (i.e., the minimum weight of an edge connecting an unvisited node to any visited nodes). The first step, where only the root node is included in the tree, is done for you.



6. (a) Analyze the worst-case runtime of QuickSort algorithm under the following assumptions.

(5+5=10)

(i) Balanced partitioning

(ii) Imbalanced partitioning

In your analysis, formulate the worst-case runtime as a recurrence relation and solve the recurrent relations.

Contd P/4

CSE 105

Contd ... Q. No. 6

- (b) Using the recurrence tree method, analyze the runtime complexity of the Mergesort algorithm to sort an array of size n . (10)
- (c) Prove that any comparison-based sorting algorithm requires $O(n \log n)$ comparisons in the worst case. (10)
- (d) Using the Master Theorem, prove that Strassen's algorithm improves the runtime of multiplying two $n \times n$ matrices compared to the naïve approach with time complexity $\theta(n^3)$. (5)
7. (a) Define the Activity Selection Problem. Prove that the greedy choice of selecting the activities based on earliest finish time yields an optimal solution. (5+10=15)
- (b) Given a graph, $G = (V, E)$ with unique edge weights, let $(S, V-S)$ be a cut in G such that S is non-empty and $S \neq V$. Show that the minimum weighted crossing edge between any such cut in G , is included in every minimum spanning tree. (10)
- (c) Sort the following list of non-negative integers using Radix Sort (using LSD - Least Significant Digit first). (5+5=10)
- [329, 457, 657, 839, 436, 720, 355]
- State the conditions required for Radix Sort to work correctly and complete in linear time.
8. (a) Given the two strings, $X = \text{"AGGTAB"}$ and $Y = \text{"GXTYAB"}$ construct the dynamic programming (DP) table to compute the length of the Longest Common Subsequence (LCS) between X and Y . Clearly state the LCS and its length. (10)
- (b) You are given a sequence of matrices $\langle A_1, A_2, \dots, A_n \rangle$ and their dimensions $\langle p_0, p_1, p_2, \dots, p_n \rangle$ such that matrix A_i has dimensions $p_{i-1} \times p_i$. The matrix-chain multiplication (MCM) problem aims to identify the minimum number of scalar multiplications necessary to determine the product of the given chain of matrices. Formulate a recurrence relation to solve this problem. Design an efficient dynamic programming algorithm using the recurrence relation. (2+8=10)
- (c) Consider the array implementation of the Union-Find data structure for some set S of size n , where union keep the name of the largest set. Analyze the run-time complexity of any sequence of k Union operations. (10)
- (d) Define stable sorting algorithm. Explain with an example. (5)